

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.3: Introduction to Electronics — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 3.3: Introduction to Electronics
Driving Question	DRIVING QUESTION: Why does a silicon solar panel on a Kajiado farm produce less power in the hot afternoon sun than on a cool morning — and what is so special about silicon that it can act as both a conductor and an insulator depending on the conditions?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Solar Panel Puzzle — Encountering the Phenomenon & Sorting Materials	Using a 1.5 V conductivity-testing circuit, we tested copper wire, plastic sufuria handle, graphite pencil lead, steel nail, rubber eraser, and a silicon chip. The LED was fully bright for copper and steel (conductors), fully off for plastic and rubber (insulators), and glowed DIMLY for graphite and silicon. This means silicon and graphite are neither full conductors nor full insulators.	Materials that conduct electricity partially — with conductivity between conductors and insulators — are called SEMICONDUCTORS. Silicon is a semiconductor. Its conductivity can change depending on conditions such as temperature and light. This is why the solar panel on Mama Akinyi's farm in Kajiado behaves differently in the morning (18 °C) versus the afternoon (35 °C).	We cannot yet fully explain WHY silicon's conductivity changes with temperature, or why this makes the solar panel less efficient when it is hotter. To explain this, we need a new model of how electrons are arranged inside solid materials — the energy band model. This will be our focus in Lesson 2.
Lesson 2 Energy Bands and the In-Between World of Silicon	In our circuit tests from Lesson 1, silicon allowed only a faint flicker of current — less than copper but more than plastic. Today we looked at the band diagram for silicon and saw that its band gap is small but not zero — about 1.1 eV. A few electrons can jump across when they receive energy from light or heat.	All solid materials have a valence band (where electrons are normally held) and a conduction band (where electrons can move freely). The size of the band gap determines conductivity. Metals have overlapping bands — no gap. Insulators have a very large gap. Semiconductors like silicon have a small gap, matched to the energy of visible light photons, which is why a solar panel made of silicon can convert sunlight into electricity.	We can now explain why silicon conducts electricity at all: sunlight provides enough energy to push electrons across the small silicon band gap into the conduction band, where they can flow as electric current. This is the first layer of the Kajiado solar panel explanation. We cannot yet explain why high afternoon temperatures reduce power output — that puzzle will be investigated in Lesson 3 using the idea of electron-hole pairs in pure silicon.
Lesson 3 Intrinsic Semiconductors — The	Learners analysed a temperature-vs-power graph for a silicon solar panel (based on the Kajiado farm phenomenon), observed that	Learners learned that solids can be classified by their energy band structure: conductors have overlapping or partially	Using the energy band model, learners can now partially explain the anchoring phenomenon: the Kajiado solar panel

Energy Band Model and Silicon's Dual Identity	power output drops as temperature rises from 18 °C to 35 °C, and noted from a band-gap data table that silicon occupies an intermediate energy gap (~1.12 eV) between conductors (0 eV) and insulators (~8 eV).	filled bands with no gap; insulators have a large band gap that electrons cannot cross at room temperature; intrinsic semiconductors like silicon have a moderate band gap (~1.12 eV) that allows a small number of electrons to be thermally promoted to the conduction band, creating electron-hole pairs. Increasing temperature increases carrier density but decreases carrier mobility due to lattice vibrations, causing a net drop in open-circuit voltage and panel power output.	produces less power on a hot afternoon (35 °C) than on a cool morning (18 °C) because, although higher temperature promotes more electrons across silicon's band gap, it simultaneously increases lattice vibrations that scatter charge carriers, reducing their mobility and lowering the panel's output voltage more than the slight current increase can compensate. Silicon's band gap of 1.12 eV gives it its semiconductor identity — it is neither a free conductor like the copper in Nairobi's power lines nor an insulator like the rubber coating those lines.
Lesson 4 Doping Silicon: How Impurities Turn a Crystal into a Switch	Colour-annotated silicon lattice diagrams showing phosphorus (donor) atoms with a free red-dot electron and boron (acceptor) atoms with a blue-circle hole; a plotted graph showing that resistivity decreases as doping concentration increases in phosphorus-doped silicon.	Doping is the deliberate addition of impurity atoms to pure silicon to control its conductivity. Phosphorus (Group V) donates a free electron, creating n-type silicon where electrons are the majority charge carriers. Boron (Group III) creates a hole (missing electron), producing p-type silicon where holes are the majority charge carriers. The higher the doping concentration, the lower the resistivity — making doped silicon far more conductive than pure (intrinsic) silicon.	The silicon in the Kajiado solar panel is not pure — it is a precisely engineered combination of n-type and p-type silicon. This doping is what allows the panel to conduct electricity in a controlled direction. In Lesson 5 we will investigate what happens at the boundary between the n-type and p-type regions (the p-n junction) to explain why current flows only one way — and why high afternoon temperatures in Kajiado reduce the panel's power output.
Lesson 5 PhET Semiconductor Lab: Exploring p-n Junctions, Bias, and Temperature Effects	In the PhET simulation, we observed that a p-n junction allows large current in one direction (forward bias) and blocks current in the other (reverse bias), producing an asymmetric I-V curve. We also observed that as temperature increased from 200K to 400K, the built-in voltage of the junction decreased and minority carrier current increased, consistent with greater thermal generation of electron-hole pairs in the doped silicon.	We learned that a p-n junction forms a depletion region with a built-in electric field that acts as a directional barrier to charge flow. In forward bias, the applied voltage reduces this barrier height, allowing exponentially increasing current. In reverse bias, the barrier is increased, blocking current. Temperature erodes the built-in voltage because more thermally generated minority carriers partially neutralise the depletion region charge, reducing the internal field strength.	We can now explain the anchoring phenomenon: on a hot Kajiado afternoon (35 degrees Celsius, approximately 308K), thermal energy generates additional electron-hole pairs in the p-n junctions of the solar panel, reducing the built-in depletion region voltage. This lowers the open-circuit voltage the panel can deliver, decreasing electrical power output — even though the number of photons hitting the panel is higher than on a cool morning. The semiconductor junction, not just the light intensity, determines how much power is produced.
Lesson 6 Case Studies — Solar Panels, Microchips & Superconducting Magnets: Applying	We analysed three pieces of evidence: (1) a power-vs-temperature graph for a silicon solar panel in Lodwar showing power output falling as panel temperature rises from 15 °C to 45 °C; (2) a diagram of a p-n junction	We learned that the same family of semiconductor properties — band gaps, doping, p-n junctions, and temperature-dependent carrier behaviour — underpins three completely different Kenyan	We can now give a more complete answer to the driving question: the Kajiado solar panel produces less power on a hot afternoon because rising temperature increases lattice vibrations in the silicon

Semiconductor Understanding to Kenyan Contexts	transistor switch inside a mobile processor showing how doping creates one-way electron flow that enables digital logic in M-PESA handsets; (3) a resistance-vs-temperature graph for a superconducting niobium-titanium alloy showing resistance dropping sharply to zero at ~9 K, as used in MRI magnets at Kenyatta National Hospital.	technologies. In solar panels, photons excite electrons across the band gap and the p-n junction drives them in one direction as useful current, but higher temperatures increase lattice vibrations that scatter carriers and reduce efficiency. In microchips, doped p-n junctions act as transistor switches that process digital signals at microscopic scales. In superconducting MRI magnets, cooling a metal alloy to near absolute zero collapses electrical resistance to exactly zero, enabling the powerful stable magnetic fields needed for medical imaging.	crystal, which scatter charge carriers more frequently, reducing their mobility and therefore the current the p-n junction can deliver. Silicon is special because its band gap is neither too large (like an insulator) nor too small (like a metal) — it can be precisely engineered through doping to create junctions that convert light into directed electron flow. This understanding, developed by semiconductor physicists, now powers rural schools in Turkana, enables financial inclusion through M-PESA, and supports life-saving medical diagnosis at KNH.
Lesson 7 Final Explanation: Semiconductors, Doping, and the Kajiado Solar Panel Mystery — Resolved	We observed that the class posters, summary tables, and video evidence from Lessons 1–6 — including the energy band diagrams, doping demonstrations, p-n junction models, and temperature-vs-efficiency data — all pointed to the same explanation. We also observed that our Lesson 1 models were incomplete: most of us thought 'more heat = more electricity,' but the evidence showed the opposite for semiconductors.	We learned that silicon is a semiconductor because its band gap is small enough for electrons to jump from the valence band to the conduction band when given energy (light or heat), but large enough to block flow in darkness. Doping silicon with phosphorus (n-type) or boron (p-type) creates a p-n junction that drives electrons in one direction when light strikes — producing a current. At higher temperatures (like 35°C on a Kajiado afternoon), increased lattice vibrations cause more electron-hole recombination before electrons can do useful work, which reduces the panel's power output. On a cooler morning (18°C), recombination is lower and efficiency is higher — explaining the farmer's observation. Semiconductor technology has transformed modern life in Kenya: M-KOPA solar home systems, Safaricom mobile network chips, and KPLC smart meters all rely on the same physics we studied.	We can now fully explain the anchoring phenomenon: The Kajiado solar panel produces less power in the hot afternoon not because there is less light, but because the higher temperature (35°C) increases lattice vibrations in the silicon crystal, raising the rate of electron-hole recombination in the p-n junction and reducing the number of electrons that complete the external circuit as useful current. On the cooler morning (18°C), recombination is lower, more photon-excited electrons cross the junction and flow through the circuit, and efficiency is higher. Silicon works as a solar panel material precisely because it is a semiconductor — its intermediate band gap allows light to excite electrons across the gap, while its p-n junction (created by doping) ensures those electrons flow in one direction as electricity. This is something neither a pure conductor (copper) nor a pure insulator (rubber) can do.

ATTRIBUTION AND LICENSING

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